

Tanmay Lad

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Education

University of Alberta

B.Sc. Computer Science, Specialization in Software Practice | Expected Apr 2028

GPA: 3.8/4.0

Relevant Coursework: Algorithms & Data Structures, Software Engineering, Database Management, Web Applications & Architecture, Advanced Games Programming, Software Process & Product Management, Foundations of Computing

Projects

Social Distribution | Django, Python, Javascript, HTML/CSS, PostgreSQL, Heroku Sept - Dec 2025

- Designed a RESTful API with 21 endpoints supporting federated interactions
- Implemented HTTP endpoints with HTTP Basic Auth and consistent HTTP status-code handling.
- Coordinated API contracts and JSON conventions with 5 development teams.
- Built automated tests for API functionality and deployed using PostgreSQL and Heroku.
- Collaborated on a 6-person development team, resolving integration and version-control issues.

PMCOLympics | Django, React, Python, TypeScript, Docker, Selenium, GitLab CI/CD Sept - Dec 2025

- Maintained and extended a React/Django web application as part of a 6-person team, working in iterative sprints with a client.
- Developed 3 new game modes, including a Wheel of Fortune game with a custom spinning-wheel animation implemented in TypeScript and CSS.
- Implemented a WebSocket-based host-peer system to support multiplayer functionality.
- Improved accessibility through responsive design and color-contrast improvements
- Used GitLab CI/CD to maintain a consistent environment and automated testing workflow.

Android Event Organizer | Android Studio, Java, XML, GitHub Jan - Apr 2024

- Developed a full stack QR-code event organizer android application using Android Studio
- Applied object-oriented design principles and Singleton, Adapter, and Command patterns to structure application components
- Practiced test-driven development (TDD) by writing automated tests using mock objects
- Collaborated through GitHub using Agile workflow to coordinate project development

StarCraft 2 Bot | C++, Blizzard API Sept - Dec 2024

- Developed a StarCraft II bot that used Blizzard's API to execute an automated strategy.
- Researched and evaluated existing StarCraft II bot designs to inform bot's implementation
- Designed and iterated the bot against automated opponents using the SC2 Ladder Server
- Applied C++ data structures including priority queues, sets, maps, and vectors

Technical Skills

- Languages: Python, C, C++, Java, JavaScript/TypeScript, HTML/CSS
- Frameworks: Django, React
- Databases: SQLite, MongoDB, Firebase, PostgreSQL
- Tools: Git, Docker, GitHub Actions, Heroku, Android Studio, Figma

